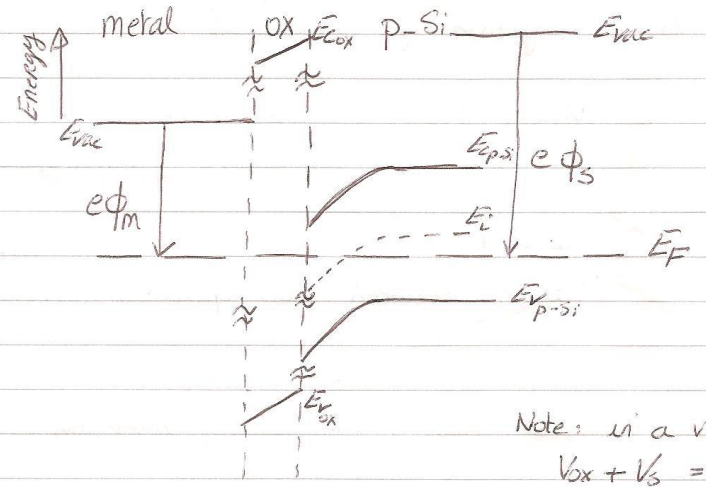
Comments to student performance in blue.

Q1

a)



[5]

Note: in a very good sketch

$V_{ox} + V_s = e(\phi_s - \phi_m)$

Comments: in general no problem

b) i) $V_s = \phi_F$ onset of depletion [1] $\Rightarrow$ $n_s = N_A \exp\left(\frac{-\phi_F}{V_T}\right)$ (1)

$n = N_c \exp\left(-\frac{(E_c - E_F)}{kT}\right)$ [2]

$n_i = N_c \exp\left(-\frac{(E_c - E_i)}{kT}\right)$

$\Rightarrow \frac{n}{n_i} = \exp\left(\frac{-E_c + E_F + E_c - E_i}{kT}\right)$

$n = n_i \exp\left(\frac{E_F - E_i}{kT}\right) = n_i \exp\left(\frac{-\phi_F}{kT}\right)$

$\Rightarrow p = p_i \exp\left(-\frac{(E_F - E_i)}{kT}\right) = p_i \exp\left(\frac{\phi_F}{kT}\right)$.

$N_A = n_i \exp\left(\frac{\phi_F}{kT}\right)$ (2)

(2) in (1): $n_s = n_i \exp\left(\frac{\phi_F}{kT}\right) \exp\left(\frac{-\phi_F}{kT}\right) = n_i$ iii)

$V_s < \phi_F$: depletion $V_s > \phi_F$: weak inversion.

p^-

$V_S = 2\phi_F \Rightarrow n_s = N_A \exp(0) = N_A$ iii) [1]

[1] ii) is the defition of threshold

$\Rightarrow$ $V_S < 2\phi_F$ weak inversion.
$V_S > 2\phi_F$ moderate inversion.

Note: both $V_S = \phi_F$ & $V_S = 2\phi_F$ define the boundary between two operation Regimes.

Comment: bookwork

c) ideal long channel MOSFET in strong inversion

$$I_{DS} = \frac{\mu C_{ox} W}{L}\left((V_{GS} - V_{th})V_{DS} - \frac{V_{DS}^2}{2}\right)$$

linear Region: $I_{DS} = \frac{\mu C_{ox} W}{L}(V_{GS} - V_{th})V_{DS}$

threshold $V_{GS} = V_{th} \Rightarrow I_{DS} = 0$

in I_{DS} equation:

$$0 = 8.85\,10^{13} V_{th} - 4.425\,10^{-13}$$
$$V_{th} = \frac{4.425}{8.85} V = 0.5V \quad [2]$$

Saturation $I_{DS} = \frac{\mu C_{ox} W}{2L}(V_{GS} - V_{th})^2 = K V_{GS}^2 - 2K V_{th} V_{GS} + V_{th}^2 K$

threshold $V_{GS} = V_{th} \Rightarrow I_{DS} = 0$

In I_{DS} equation:

$$0 = 4.425\,10^{-11} V_{th}^2 - 3.9825\,10^{-11} V_{th} + 8.9606\,10^{-12}$$
$$0 = 4.425\,V_{th}^2 - 3.9825 V_{th} + 0.8960625$$

$$V_{th} = \frac{+3.9825 \pm \sqrt{(3.9825)^2 - 4 \times 4.425 \times 0.8960625}}{2 \times 4.425}$$

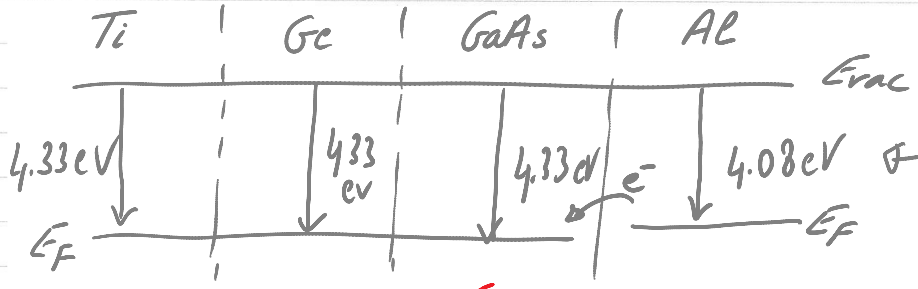
$= 0.45V$ [2]

Comment: this was easy once
1°: correct formulae for I_{DS}^{lin} & I_{DS}^{sat} were written
2°: factor in front of V_{GS} resp V_{GS}^2 is divided away.

$$DIBL = \left| \frac{V_{th}^{lin} - V_{th}^{sat}}{V_{DS}^{lin} - V_{DS}^{sat}} \right| = \frac{0.5 - 0.45}{|0.01 - 1|} = 0.051 \frac{V}{V}$$ [1]

$$= 51 \frac{mV}{V}$$

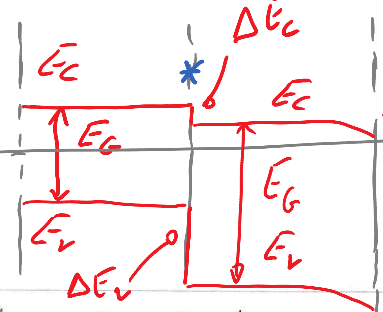
d)



Φ workfunctions

Comment: most errors happen @ *. Since $\Phi_{Ge} = \Phi_{GaAs}$ => no band bending but since $\chi_{Ge} < \chi_{GaAs}$ [5] $\Delta E_c \neq 0$ & $E_{c_{GaAs}}$ lower than $E_{c_{Ge}}$.

Ge
$E_G = 0.66eV$
$\Delta E_c = 4 - 4.1 = 0.1eV$
GaAs
$E_G = 1.42eV$
$\Delta E_v = \Delta E_G - \Delta E_c$
$= 0.66eV$



e) i) $\mu_1 = \mu_2 < \mu_3$

OR $\mu_3 > \mu_2 = \mu_1$ [3]

slope @ low electric field is the same.

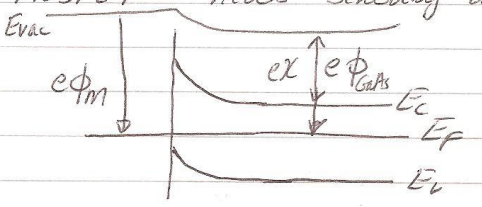
ii) $v_{sat_2} < v_{sat_3} < v_{sat_1}$

OR

$v_{sat_1} > v_{sat_3} > v_{sat_2}$ [2]

$v_{saturation}$ @ high electric fields only.

f) n-type MESFET needs Schottky contact.



$\Phi_{GaAs} < \Phi_m$ for Schottky contact.

[5]

$$\Phi_{GaAs} = \chi_{GaAs} + \frac{(E_c - E_F)}{C} \qquad N_D = 10^{15} cm^{-3} \rightarrow \text{very low}$$

$$\Phi_{GaAs} = 4.1\,eV + 0.156\,eV \cong 4.26\,eV$$

$$n = N_c \exp\left(\frac{-(E_c - E_F)}{kT}\right)$$

$$E_c - E_F = kT \ln\left(\frac{N_c}{n}\right)$$

$$= 0.026 \ln\left(\frac{4.7\,10^{17}}{10^{15}}\right) = 0.156\,eV$$

N_i will be the best theoretical choices because this gives the largest $V_{bi} = \Phi_m - \Phi_{GaAs}$.

Comment: asked was: "the best", this means that only N_i is ok.

g) $I_{0G}^{MESFET} > I_{0G}^{JFET}$

the gate of a MESFET is determined by a Schottky barrier, that of a JFET by a pn-junction.

the leakage current through the gate is related to this barrier height.
Due to interface charges & traps in a metal/semiconductor contact the barrier height in a MESFET is lower than in a JFET [5]

$$\Rightarrow I_{G0}^{MESFET} > I_{G0}^{JFET}.$$

Comment: the important info here is the magnitude of the barrier height and the fact that MESFET gates suffer from interface states which causes the barrier to remain low whatever the chosen metal.

h) i) fin FET [1]

ii) a) gate
b) buried oxide layer
c) gate oxide.
d) Si substrate. [4]

↳ bookwork
it is important to distinguish "oxide" in b) & c)

(2) a) Small drain voltage = linear (triode) region [8]

W_{max} = maximum depletion width from gate = @ threshold

→ $W_{max} = W_{depl}(V_S = 2\phi_F)$

formulae sheet: * $W_{depl} = \left[\frac{2\varepsilon_0\varepsilon_{Si}(V_{bi}-V)}{e}\frac{1}{N_A}\right]^{1/2}$ for 1-sided junction

$V_{bi} = 0$: only consider V_S and anyway $\phi_m = \phi_{p\text{-}Si}$

$$W_{max} = \sqrt{\frac{2\,\varepsilon_0\,\varepsilon_S}{e\,N_A}\,2\,\frac{kT}{e}\ln\left(\frac{N_A}{n_i}\right)}$$

$$W_{max} = \sqrt{\frac{4\times 8.85\,10^{-14}\,\frac{F}{cm}\,12\,0.026\,\ln\left(\frac{10^{16}}{1.45\,10^{10}}\right)}{1.6\,10^{-19}\,C\;10^{16}\,cm^{-3}}} = 3.05\,10^{-5}\,cm$$

Comment: main error occurs with the choice of V. $V = -V_S$, the surface potential only. V_{th} includes V_{ox}! Since $V_S = 2\phi_F$ @ $W_{depl\,max}$ V_{th} is not needed.

* $W_{DB} = \sqrt{\frac{2\,\varepsilon_0\varepsilon_{Si}\,(V_{bi}+0.26)}{e\,N_A}}$ assume depletion region extends in the lowest doping side only.

Comment: the value ε_0 is given in F/m in formulae sheet but doping densities are in cm^{-3} ⇒ many unit errors

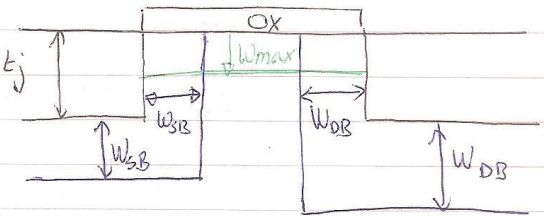
$$V_{bi} = \frac{kT}{e}\ln\left(\frac{N_A N_D}{n_i^2}\right) = 0.026\ln\left(\frac{10^{16}\,10^{20}}{(1.45\,10^{10})^2}\right) = 0.94\,V$$

$$W_{DB} = \sqrt{\frac{2\;8.85\,10^{-14}\;12\;(0.94+0.26)}{1.6\,10^{-19}\;10^{16}}} = 3.99\,10^{-5}\,cm$$

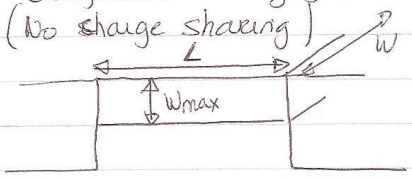
* W_{SB} : same assumption as for W_{DB} and $V_S = 0\,V$

$$W_{SB} = \sqrt{\frac{2\times 8.85\,10^{-14}\;12\;0.94}{1.6\,10^{-19}\;10^{16}}} = 3.53\,10^{-5}\,cm$$

b) [4]



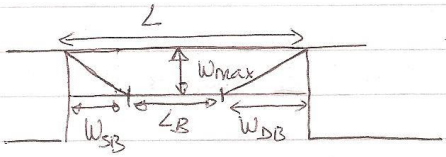
c) Q_G (No charge sharing) — Charge controlled by gate when



[8]

$$Q_G = \frac{-e\, N_A\, W_{max}\, L\, W}{LW} \quad : \quad \text{(charge per gate area)}$$

Q_{GCS} (With charge sharing) — charge controlled by gate



area trapezium : $\left(\frac{L+L_B}{2}\right) \cdot W_{max}$

$$Q_{GCS} = \frac{-e\, N_A\, W_{max}}{LW}\left(\frac{L+L_B}{2}\right) W$$

$$= \frac{-e\, N_A\, W_{max}}{LW}\left(\frac{L + (L - W_{SB} - W_{DB})}{2}\right) W$$

charge per gate area.

charge lost : $\Delta Q = Q_G - Q_{GCS}$

$$= \frac{-eN_A\, W_{max}\, W}{LW}\left(L - \left(\frac{2L - W_{SB} - W_{DB}}{2}\right)\right)$$

$$= \frac{-e\, N_A\, W_{max}\, W}{LW}\left(\frac{2L - 2L + W_{SB} + W_{DB}}{2}\right)$$

$$= \frac{-e\, N_A\, W_{max}\, W}{LW}\left(\frac{W_{SB} + W_{DB}}{2}\right)$$

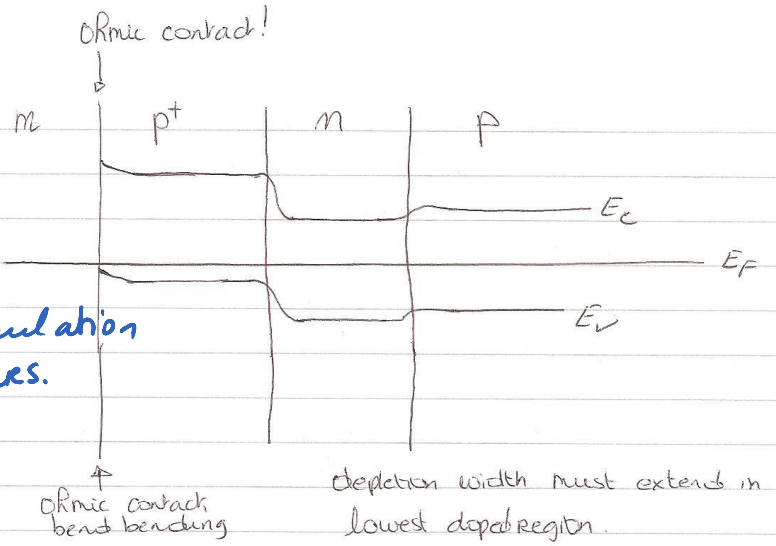
$$\Delta V_{th} = \frac{-e\, N_A\, W_{max}}{2\, C_{ox}\, L}\left(W_{SB} + W_{DB}\right)$$

Note : signs are important because :
1° charge is negative
2° V_{th} decreases (threshold voltage roll off)

Comment: depletion widths often not drawn consistent with doping density. thus junctions & depletion width edges should be given in answer.

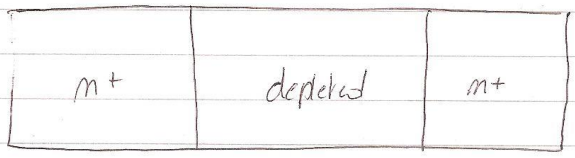
3. a)

Comment: this is a JFET ⇒ the n-p+ junction should be Ohmic!! thus or flat band or accumulation of majority carriers.



[5]

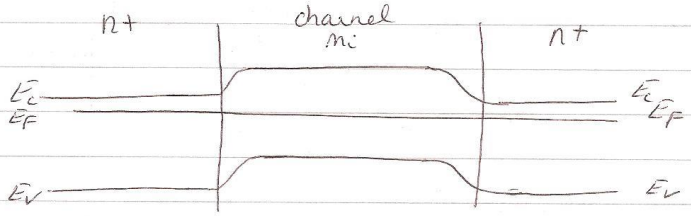
b)



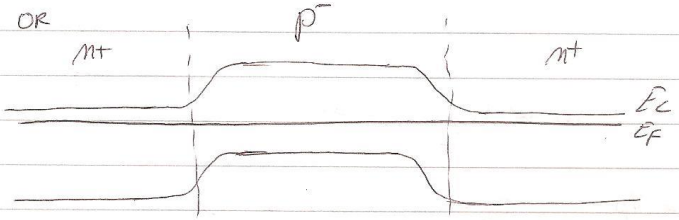
[5]

Note, in the MOSFET part of the course, we have seen that depleted means that the electron concentration ≦ n_i

⇒ possibilities



OR



* p^+ in channel region is not acceptable because that would require to high a V_{GS}
what is acceptable are plots of n^+ n n_i n n^+ or n^+ n p^- n n^+
again extend of depletion width is important.

c) Pinch-off occurs when the channel is depleted, ~~&~~
Since it is stated that the differences in doping concentration need to be taken into account, the depletion extending from the p^+-type region & that from the bulk both need to be taken into account.

[10]

Depletion from bulk:

W_B

channel $N_D = 10^{17}\ cm^{-3}$
bulk $N_A = 5 \cdot 10^{17}\ cm^{-3}$
} only 5x different ⇒ have to take both into account

In formulae sheet $W = \sqrt{\frac{2\varepsilon_0 \varepsilon_{Si}(V_0 - V)}{e}\left(\frac{1}{N_A} + \frac{1}{N_D}\right)}$

we only need W_p

we know that $W_n + W_p = W$ (1)

charge neutrality $W_n N_D = W_p N_A$ (2)

(2) $W_n = W_p \frac{N_A}{N_D}$ (3)

(3) in (1) $W_p\left(\frac{N_A}{N_D} + 1\right) = W$

$W_p\left(\frac{N_A + N_D}{N_D}\right) = W$

$V_B = 0 \rightarrow W_p = \sqrt{\frac{2\varepsilon_0 \varepsilon_s}{e}(V_{0_B})\left(\frac{N_A + N_D}{N_A N_D}\right)}\left(\frac{N_D}{N_A + N_D}\right)$

$W_p = W_{B_p} = \sqrt{\frac{2\varepsilon_0 \varepsilon_s V_{0_B}}{e}\left(\frac{N_D}{N_A(N_A + N_D)}\right)}$

Depletion from gate (p^+)

gate p^+ : $10^{20}\ cm^{-3}$
channel n : $10^{17}\ cm^{-3}$
} since $N_A^+ >> N_D^-$ assume $W_{p+} << W_{n-}$ ⇒ 1-sided junction.

$W \cong W_{p_G} = \sqrt{\frac{2\varepsilon_0 \varepsilon_{Si}(V_{0_G} - V_{GS})}{e N_D}}$

@ $V_{GS} = V_p$ $\quad t_n - W_{B_p} - W_{P_G} = 0$

$W_{P_G} = t_n - W_{B_p}$

Comment: This question was poorly answered. Most answers ignored the depletion from the bulk. It was asked to take doping differences into account ⇒ since doping in channel & substrate are ± same, the depletion should be taken into account. The other error was that the total depletion width was taken into account, rather than the depletion width extending into the channel only.

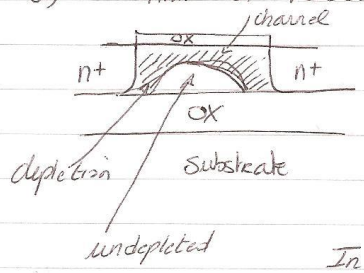
Other comments: * this is a JFET not a MOSFET ⇒ different approach.
* too much reproduction from course notes which is not what is asked for in this question.

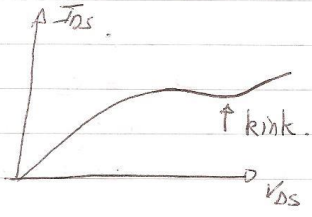
$$\sqrt{\frac{2\varepsilon_0\varepsilon_{Si}(V_{OG}-V_P)}{e N_D}} = t_n - \sqrt{\frac{2\varepsilon_0\varepsilon_S}{e} V_{OB}\left(\frac{N_D}{N_{A_B}(N_{A_B}+N_D)}\right)}$$

$$\frac{2\varepsilon_0\varepsilon_{Si}(V_{OG}-V_P)}{e N_D} = \left(t_n - \sqrt{\frac{2\varepsilon_0\varepsilon_S}{e} V_{OB}\left(\frac{N_D}{N_{A_B}(N_{A_B}+N_D)}\right)}\right)^2$$

$$V_P = V_{OG} - \frac{eN_D}{2\varepsilon_0\varepsilon_{Si}}\left(t_n - \sqrt{\frac{2\varepsilon_0\varepsilon_{Si}}{e} V_{OB}\left(\frac{N_D}{N_{A_B}(N_{A_S}+N_D)}\right)}\right)^2$$

(4) a) kink in PDSOI





[5]

In PDSOI, the depletion region from the gate does not reach the BOX → there is a region of undepleted Si. When the voltage V_{DS} increases, impact ionisation occures @ drain creating e^-R^+ pairs. e^- get extracted in D but R^+ build up in the undepleted region. This causes a change in charge in that region → thus change in Q_{depl} and thus a change in $\Delta V_{th} = \frac{\Delta Q_{depl}}{C_{ox}}$

V_{th} ↓ ⇒ [redacted] ⇒ I_{DS} ↑ → kink

b) advantages: ⊛ isolation between nMOS & pMOS Removes latch-up [5]

(take 3 out of) ⊛ fewer processing steps needed

- (*) Reduce substrate leakage ⇒ lower power consumption.
- (*) Higher packing density
- (*) smaller parasitic capacitances ⇒ higher speed

disadvantages.

(take 2 out of)

- (*) Completely surrounded by oxide ⇒ heating problem
- (*) SOI substrate more expensive
- (*) transients : minority carrier {recombination / generation} via {depletion / inversion} supply only available

c) for fully depleted SOI t_c needs to be depleted @ $V_{GS} = 0V$ ⇒ the depletion region from the gate $d_G = t_c$. This depletion region can only be formed by the workfunction difference between n-Si & ~~metal~~ gate contact.

from formulae list:

$$W_{depl} = \left[\frac{2\varepsilon (V_{bi} - V)}{e} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) \right]^{1/2}$$

{one-sided junction ⇒ $V_{GS} = 0V$

$$W_{depl} = \left[\frac{2\varepsilon V_{bs}}{e N_A} \right]^{1/2}$$

$V_{bi} = 0$ but $V_S \neq 0$ [10]

$W_{depl\,max}$ when $V_S = 2\phi_F$

the maximum width of t_c is when $V_{th} = 0V$

thus $t_c \leqslant W_{depl}\,(V_{th} = 0V)$

V_{th} from formulae list:

$$V_{th} = \phi_m - \phi_S + 2\phi_F + \gamma \sqrt{2\phi_F}$$

$$\gamma = \frac{\sqrt{2e\varepsilon_S N_A}}{C_{ox}}$$

$$V_{th} = 0 = \phi_m - \phi_s + 2\phi_F + \gamma\sqrt{2\phi_F}$$

$$\Rightarrow \phi_m = \phi_s - 2\phi_F - \gamma\sqrt{2\phi_F}$$

Comment:
$V_{bi} = 0$
$V_s \neq 0$: draw energy band diagram!
The voltage for $W_{depl}{}_{max}$ is not $\phi_m - \phi_{si}$ because that ignores that part of this potential difference is dropped across the oxide:
$V_{ox} + V_s = \phi_m - \phi_{si}$
Note that $\phi_m < \phi_{si}$ in order for this to work => signs are important.

V_{bi} in $W_{depl} = ?$

$V_s = 2\phi_F$ when max depl. region occurs.

$$t_c^{max} = \sqrt{\frac{2\varepsilon\, 2\phi_F}{eN_A}} \quad \text{with } \phi_F = V_T \ln\left(\frac{N_A}{n_i}\right)$$

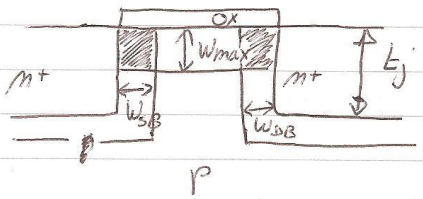
$$\Rightarrow t_{c_{max}}^2 = \frac{4\varepsilon\phi_F}{eN_A}$$

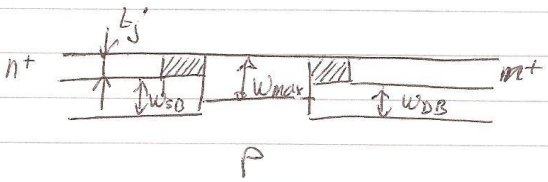
$$\phi_F = \frac{eN_A t_{c_{max}}^2}{4\varepsilon}$$

$$\phi_m = \phi_s - \frac{eN_A t_{c_{max}}^2}{2\varepsilon_0\varepsilon_{si}} - \gamma\sqrt{\frac{eN_A t_{c_{max}}^2}{2\varepsilon_0\varepsilon_{si}}}$$

$\hookrightarrow \phi_m^{min}$ = minimum value of ϕ_m.

(5)





$V_{DS} = 0$ $V_{GS} = V_{th}$

W_{max} : max gate induced depletion width

W_{SB}/W_{DB} : S & D depletion width

▨ area of shaked charge ✓

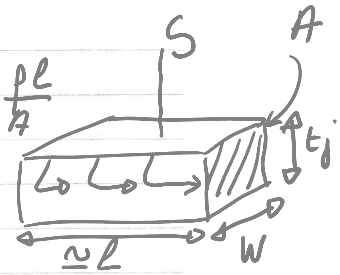
[5]

▨ area of shaked charge t_j ✓

all other parameters same.

b) $t_j \downarrow \Rightarrow$ contact resistance $\nearrow$

$\Rightarrow R_S \nearrow$ (source resistance) $\quad R_S = \frac{\rho l}{A}$



$$g_m = \frac{dI_{DS}}{dV_{GS}} \quad \text{or} \quad \frac{1}{g_m} = \frac{dV_{GS}}{dI_{DS}}$$

$$V_{GS}^{ext.} = V_{GS}^{int} + R_S I_{DS} \qquad \text{assume} \begin{cases} I_G = 0\,V \\ I_{bulk} = 0V \end{cases}$$

$$\frac{1}{g_m^{ext}} = \frac{dV_{GS}^{ext}}{dI_{DS}}$$

$$= \frac{d\left(V_{GS}^{int} + R_S I_{DS}\right)}{dI_{DS}} = \frac{dV_{GS}^{int}}{dI_{DS}} + R_S$$

$$\frac{1}{g_m^{ext}} = \frac{1}{g_m^{int}} + R_S \qquad [8]$$

$$\frac{1}{g_m^{ext}} = \frac{1 + R_S g_m^{int}}{g_m^{int}}$$

$$g_m^{ext} = \frac{g_m^{int}}{1 + R_S \cdot g_m^{int}}$$

ext : externally measured g_m

int : internally " "

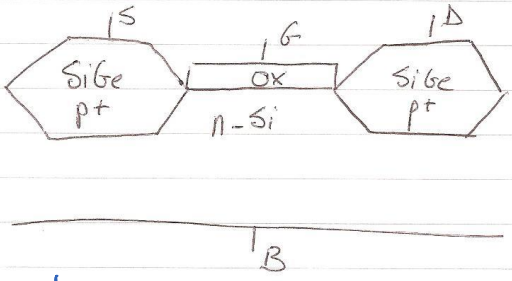
(without R_S parasitic)

c) Comment : important here is to derive g_m^{ext} as a function of g_m^{int} & R_S and to define what "A" is for R_S.

c) i) elevated source-drain technology. [2]

ii) SiGe has a larger lattice constant than Si, thus when the Si channel is surrounded by SiGe (e.g. in the source & drain areas) it puts Si under compressive strain increasing hole mobility (for pMOS). [2]

iii)



[3]

bookwork. one has to make the connection between elevated source-drain & local strain.